Unit 3

Module 3 : Ordinary Differential Equations of Higher Orders

4

Series Solution of Linear Ordinary Differential Equations

Contents

Syllabus

Classification of Ordinary and Singular points, Power series solutions for ordinary points.

4.1 Introduction

In previous chapter, we studied different method to solve linear ordinary differential equation with constant co-efficient. In this chapter, we studied linear ordinary differential equation with variable co-efficient. This method can be used to solve where Euler Cauchy's equation with variable co-efficient is not applicable.

The power series method gives the solution in the form of the convergent infinite series. This series solution of linear differential equation with variable co-efficient can be used for computing valves, ploting graphs etc.

4.2 Basic Concept

4.2.1 Power Series

An infinite series of the form $\sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \ldots + a_n(x-x_0)^n + \ldots$ is said to be a power series in the power of $(x-x_0)$, where $a_0, a_1, a_2, \ldots, a_n$ are co-efficient of a series and x_0 is called center.

If we consider center $x_0 = 0$ then series of the form $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n + \ldots$ is called power series in the power of x.

- **Analytic function :** A function f(x) defined on an interval containing the point $x = x_0$ is called analytic at x_0 if its Taylor series exist and converges to f(x) for all x in the interval of convergence.
- **Remark 1 :** Every polynomial function, exponential function, sine function, cosine function are analytic everywhere.
- **Remark 2 :** Any ration function like, $\frac{f(x)}{g(x)}$ is not analytic at those values of x, where denominator

  $g(x) = 0$.

Example : $f(x) = x^2 + 2x + 1$, e^{2x}, cos 3x, sin x are analytic everywhere but $f(x) = \frac{x^2+2}{x-2}$ is not analytic at x = 2. $f(x) = \frac{x^3+1}{(x-1)(x-2)(x-3)}$ is not analytic at x = 1, x = 2 and x = 3.

4.3 Classification of Ordinary and Singular Points

GTU : Summer-19,20,23,24, Winter-19,21,22,23, Marks 7

Let us consider a 2^{nd} order linear ordinary differential equation with variable co-efficient is,

$$a(x) \cdot \frac{d^2y}{dx^2} + b(x) \cdot \frac{dy}{dx} + c(x) \cdot y = 0 \quad \ldots(4.3.1)$$

where, a(x), b(x) and c(x) are analytic function of x.

Now equation (4.3.1) can be written as,

$$\frac{d^2y}{dx^2} + \frac{b(x)}{a(x)} \cdot \frac{dy}{dx} + \frac{c(x)}{a(x)} \cdot y = 0$$

Let, $$P(x) = \frac{b(x)}{a(x)} \quad \text{and} \quad Q(x) = \frac{c(x)}{a(x)}$$

$$\frac{d^2y}{dx^2}+P(x)\cdot\frac{dy}{dx}+Q(x)\cdot y = 0 \qquad ...(4.3.2)$$

∴

- If $a(x) \neq 0$ at $x = x_0$ then $x = x_0$ is said to be a ordinary points. (where, P(x) and Q(x) analytic at $x = x_0$)
- If $a(x) = 0$ at $x = x_0$ means P(x) or Q(x) is not analytic at $x = x_0$ then $x = x_0$ is said to be a singular points.
- In case of $x = x_0$ singular points, then $x = x_0$ can be regular singular or irregular signular points.
  i) If $(x-x_0)P(x)$ and $(x-x_0)^2Q(x)$ are analytic at $x = x_0$ then $x = x_0$ is said to be a regular singular points.
  ii) If $(x-x_0)P(x)$ or $(x-x_0)^2Q(x)$ or both are not analytic at $x = x_0$ then $x = x_0$ is said to be a irregular singular points.

Ex. 4.3.1 : *Classify the signularity for the following differential equation.*

i) $y''+3x^2y'+2y = 0$
ii) $y''+x^2y+xy = 0$
iii) $(x^2+1)\,y''+x^2y'+xy = 0$
iv) $(1-x^2)\,y''-2xy'+y = 0$
v) $x^2y''+xy'-2y = 0$
vi) $x^3(x-1)y''+3(x-1)y'+xy = 0$

GTU : Winter-23, AEM, Marks 4

Sol. : i) $y''+3x^2y'+2y = 0$ is given

We compare with standard form.

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \qquad P(x) = 3x^2 \text{ and } Q(x) = 2$$

Here, P(x) and Q(x) both are analytic everywhere.

∴ All points are ordinary.

ii) Here, $y''+x^2y+xy = 0$ is given

Compare with standard form.

$$y''+P(x)\cdot y+Q(x)\cdot y = 0$$

$$\therefore \qquad P(x) = x^2 \text{ and } Q(x) = x$$

Here, P(x) and Q(x) both analytic everywhere

∴ All points are ordinary.

iii) $(x^2+1)\,y''+x^2y'+xy = 0$ is given

First we write in the form.

$$y''+\frac{x^2}{1+x^2}y'+\frac{x}{1+x^2}y = 0$$

Now compare with $y''+P(x)\cdot y'+a(x)\cdot y = 0$

$$\therefore \quad P(x) = \frac{x^2}{1+x^2} \quad \text{and} \quad Q(x) = \frac{x}{1+x^2}$$

Here, P(x) and Q(x) is analytic everywhere for all x, $x \in R$.

∴ All points are ordinary.

iv) Here, $(1-x^2)\,y''-2xy'+y = 0$ is given

First, we write (Divide with $(1-x^2)$)

$$y''-\frac{2x}{(1-x^2)}y'+\frac{1}{(1-x^2)}y = 0 \quad ...(A)$$

Equation (A) compare with standard form.

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \qquad P(x) = \frac{-2x}{1-x^2}, \quad Q(x) = \frac{1}{1-x^2}$$

$$= \frac{-2x}{(1-x)(1+x)} = \frac{1}{(1-x)(1+x)}$$

Here, P(x) and Q(x) are not analytic at $x = 1$ and $x = -1$.

(∵ Take denominator equal to zero

$$\therefore \quad (1-x^2) = 0 \Rightarrow (1-x)(1+x) = 0$$

$$\Rightarrow x = 1, x = -1)$$

∴ $x = 1$ and $x = -1$ are singular points.

All points are ordinary except $x = 1$ and $x = -1$

Now, we check singular points are which type regular singular or irregular singular.

For $x = 1$

$$(x-1)\,P(x) = \frac{-2x(x-1)}{(x-1)(x+1)} = \frac{-2x}{x+1}$$

$$(x-1)^2Q(x) = \frac{-2x(x-1)^2}{(x-1)(x+1)} = \frac{-2x(x-1)}{x+1}$$

Here, $(x-1)$ P(x) and $(x-1)^2Q(x)$ both analytic at $x = 1$.

∴ $x = 1$ is regular singular point.

For $x = -1$

$$(x+1)P(x) = \frac{-2x(x+1)}{(x-1)(x+1)} = \frac{-2x}{x-1}$$

$$(x+1)^2 Q(x) = \frac{-2x(x+1)^2}{(x-1)(x+1)} = \frac{-2x(x+1)}{(x-1)}$$

Here, $(x+1)P(x)$ and $(x-1)^2 Q(x)$ both analytic at $x = -1$.

$\therefore$ $x = -1$ is regular singular point.

v) Here, $x^2y'' + xy' - 2y = 0$ is given

GTU : Summer-20, Maths-2, Marks 3

First, we write (we divide x^2)

$$y'' + \frac{1}{x}y' - \frac{2}{x^2}y = 0$$

Compare with standard form.

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = \frac{1}{x} \quad \text{and} \quad Q(x) = \frac{-2}{x^2}$$

Here, P(x) and Q(x) are not analytic at $x = 0$.

$\therefore$ $x = 0$ is singular point.

Now, we check $x = 0$ is regular singular or irregular singular.

For $x = 0$. We multiply $(x - 0)$ with P(x) and $(x-0)^2$ with Q(x).

$$\therefore \quad (x-0)P(x) = \frac{(x-0)}{x} = 1$$

$$\therefore \quad (x-0)^2 Q(x) = \frac{-2(x-0)^2}{x^2} = -2$$

$\therefore$ $(x-0)P(x)$ and $(x-0)^2Q(x)$ both analytic at $x = 0$

$\therefore$ $x = 0$ is regular singular point.

vi) Here, $x^3(x-1)y'' + 3(x-1)y' + xy = 0$ is given.

First, we write (we divide $x^3(x-1)$)

$$y'' + \frac{3(x-1)}{x^3(x-1)}y' + \frac{x}{x^3(x-1)}y = 0$$

$$\therefore \quad y'' + \frac{3}{x^2}y' + \frac{1}{x^2(x-1)}y = 0$$

Compare with standard form.

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = \frac{3}{x^2} \quad \text{and} \quad Q(x) = \frac{1}{x^2(x-1)}$$

Here, P(x) is not anlaytic at $x = 0$ and Q(x) is not analytic at $x = 0$ and $x = 1$.

For, $x = 0$ we multiply $(x - 0)$ with P(x) and $(x-0)^2$ with Q(x).

$\therefore$ $x = 0$ and $x = 1$ is singular points

Now we check $x = 0$ and $x = 1$ are regular singular or irregular singular.

$$\therefore \quad (x-0)P(x) = \frac{3(x-0)}{x^2} = \frac{3}{x}$$

$$(x-0)^2 Q(x) = \frac{(x-0)^2}{x^2(x-1)} = \frac{1}{x-1}$$

Here, $(x - 0)P(x)$ again not analytic at $x = 0$.

$\therefore$ $x = 0$ is irregular singular point.

For, $x = 1$, we multiply $(x - 1)$ with P(x) and $(x-1)^2$ with Q(x).

$$\therefore \quad (x-1)P(x) = \frac{3(x-1)}{x^2} = \frac{3(x-1)}{x^2}$$

$$(x-1)^2 Q(x) = \frac{(x-1)^2}{x^2(x-1)} = \frac{(x-1)}{x^2}$$

Here, $(x-1)P(x)$ and $(x-1)^2Q(x)$ is analytic at $x = 1$

$\therefore$ $x = 1$ is regular singular point.

Ex. 4.3.2 : *Find the ordinary points of $(x^2+1)y'' + 2xy' + 6y = 0$.*

GTU : Winter-22, Maths-2, Marks 3

Sol. : Here, $(x^2+1)y'' + 2xy' + 6y = 0$ is given

First, we write (Divide by (x^2+1))

$$y'' + \frac{2x}{1+x^2}y' + \frac{6}{1+x^2}y = 0$$

$$\therefore \quad P(x) = \frac{2x}{1+x^2}, \quad Q(x) = \frac{6}{1+x^2}$$

Here, P(x) and Q(x) both are analytic everywhere, for all points of $x \in R$.

$\therefore$ All points are ordinary.

Ex. 4.3.3 : *Determine the singular points of the differential equation $2x\,(x-2)^2 y'' + 3xy' + (x-2)\,y = 0$ and classify them as regular or irregular.*

GTU : Winter-23, Maths-2, Marks 3

Sol. : Here, $2x(x-2)^2 y'' + 3xy' + (x-2)y = 0$ is given.

First, we write (Divide $2x(x-2)^2$)

$$\therefore \quad y'' + \frac{3x}{2x(x-2)^2} y' + \frac{(x-2)}{2x(x-2)^2} y = 0$$

$$\therefore \quad P(x) = \frac{3}{2(x-2)^2} \quad Q(x) = \frac{1}{2x(x-2)}$$

Here, P(x) is not analytic at x = 2 and Q(x) is not analytic at x = 0 and x = 2.

$\therefore$ x = 0 and x = 2 is singular points.

Now, we check x = 0 and x = 2 is regular singular or irregular singular.

For x = 0, we multiply (x – 0) with P(x) and $(x-0)^2$ with Q(x).

$$\therefore \quad (x-0)\,P(x) = \frac{(x-0)\cdot 3}{2(x-2)^2} = \frac{3x}{2(x-2)^2}$$

$$Q(x)(x-0)^2 = \frac{(x-0)^2}{2x(x-2)} = \frac{x}{2(x-2)}$$

Here, (x – 0) P(x) and $(x-0)^2 Q(x)$ are also analytic at x = 0

$\therefore$ x = 0 is regular singular point.

For x = 2, we multiply P(x) with (x – 2) and Q(x) with $(x-2)^2$

$$(x-2)\,P(x) = \frac{(x-2)\cdot 3}{2(x-2)^2} = \frac{3}{2}\cdot\frac{1}{(x-2)}$$

$$(x-2)^2 Q(x) = \frac{(x-2)^2}{2x(x-2)} = \frac{(x-2)}{2x}$$

Here, (x – 2) P(x) is not analytic at x = 2.

$\therefore$ x = 2 is irregular singular point.

Ex. 4.3.4 : *Discuss about the ordinary point, regular singular points and irregular singular points for the differential equation $x^3 y'' + 5xy' + 3y = 0$*

GTU : Winter-23, Maths-2, Marks 3

Sol. : Here, $x^3 y'' + 5xy' + 3y = 0$ is given

First, we write (Divide by x^3)

$$y'' + \frac{5}{x^2} y' + \frac{3}{x^3} y = 0$$

Compare with standard form.

$$y'' + P(x)\cdot y' + Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = \frac{5}{x^2}, \quad Q(x) = \frac{3}{x^3}$$

Here, P(x) and Q(x) is analytic everywhere except x = 0.

$\therefore$ x = 0 is singular point and other points are ordinary.

Now, we check x = 0 is regular or irregular.

For, x = 0. We multiply P(x) with (x – 0) and Q(x) with $(x-0)^2$.

$$\therefore \quad (x-0)\,P(x) = \frac{(x-0)}{x^2} = \frac{1}{x}$$

$$(x-0)^2 Q(x) = \frac{(x-0)^2\cdot 3}{x^3} = \frac{3}{x}$$

Here, (x – 0) P(x) and $(x-0)^2 Q(x)$ is again not analytic at x = 0

$\therefore$ x = 0 is irregular singular point.

Ex. 4.3.5 : *Find the ordinary and singular points of $2x^2 y'' + 6xy' + (x+3)y = 0$.*

GTU : Winter-21, Maths-2, Marks 4

Sol. : Here, $2x^2 y'' + 6xy' + (x+3)y = 0$ is given.

First, we write (Divide by $2x^2$)

$$\therefore \quad y'' + \frac{3}{x} y' + \frac{(x+3)}{2x^2} y = 0$$

Compare with standard form.

$$y'' + P(x)\cdot y' + Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = \frac{3}{x} \quad \text{and} \quad Q(x) = \frac{(x+3)}{2x^2}$$

Here, P(x) and Q(x) is analytic every points except x = 0.

$\therefore$ All points are ordinary except x = 0 and x = 0 is singular point.

Ex. 4.3.6 : *Classify the singular points of the equation* $x^3(x-2)y''+x^3y'+6y=0$

GTU : Winter-19, Summer-24, Maths-2, Marks 3

Sol. : Here, $x^3(x-2)y''+x^3y'+6y=0$ is given.

First we write (Divide by $x^3(x-2)$)

$$y''+\frac{1}{x-2}y'+\frac{6}{x^3(x-2)}y = 0$$

Compare with standard form.

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = \frac{1}{x-2} \quad Q(x) = \frac{6}{x^3(x-2)}$$

Here, P(x) is not analytic at x = 2

and Q(x) is not analytic at x = 0 and x = 2

$\therefore$ x = 0 and x = 2 is singular points and other points are ordinary.

Now, we check x = 0 and x = 2 are regular singular or irregular singular.

For x = 0

$$(x-0)\,P(x) = \frac{(x-0)\cdot 1}{x-2} = \frac{x}{x-2}$$

$$(x-0)^2Q(x) = \frac{(x-0)^2\cdot 6}{x^3(x-2)} = \frac{6}{x(x-2)}$$

Here, $(x-0)^2Q(x)$ again not analytic at x = 0

$\therefore$ *x = 0 is irregular singular.*

For x = 2

$$(x-2)\,P(x) = \frac{(x-2)\cdot 1}{(x-2)} = 1$$

$$Q(x)(x-2)^2 = \frac{(x-2)^2\cdot 6}{x^3(x-2)} = \frac{6(x-2)}{x^3}$$

Here, (x − 2) P(x) and $(x-2)^2Q(x)$ both analytic at x = 2.

$\therefore$ x = 2 is regular singular point.

Ex. 4.3.7 : *Discuss about ordinary point, singular point, regular singular point and irregular singular point at differential equation* $(x^2-1)y''+3xy'+5x^2y=0$

GTU : Summer-23, Maths-2, Marks 3

Sol. : Here, $(x^2-1)y''+3xy'+5x^2y=0$ is given.

First, we write (Divide by (x^2-1))

$$y''+\frac{3x}{x^2-1}y'+\frac{5x^2}{x^2-1}y = 0$$

Compare standard equation,

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = \frac{3x}{x^2-1} = \frac{3x}{(x-1)(x+1)}$$

$$Q(x) = \frac{5x^2}{x^2-1} = \frac{5x^2}{(x-1)(x+1)}$$

Here, P(x) and Q(x) are not analytic at x = 1 and x = − 1.

$\therefore$ Both points x = 1 and x = − 1 are singular and other points are ordinary.

Now, we check x = 1 and x = − 1 are regular singular or irregular singular.

For x = 1

$$(x-1)\,P(x) = \frac{3x\cdot(x-1)}{(x-1)(x+1)} = \frac{3x}{(x+1)}$$

$$(x-1)^2Q(x) = \frac{(x-1)^2\cdot 5x^2}{(x-1)(x+1)} = \frac{(x-1)\cdot 5x^2}{x+1}$$

Here, (x − 1) P(x) and $(x-1)^2Q(x)$ both analytic at x = 1.

$\therefore$ x = 1 is regular singular point.

For x = − 1

$$\therefore \quad (x+1)\,P(x) = \frac{3x(x+1)}{(x-1)(x+1)} = \frac{3x}{x-1}$$

$$(x+1)^2Q(x) = \frac{(x+1)^2\cdot 5x^2}{(x-1)(x+1)} = \frac{(x+1)\cdot 5x^2}{(x-1)}$$

Here, (x + 1) P(x) and $(x+1)^2Q(x)$ both analytic at x = − 1.

$\therefore$ x = − 1 is regular singular point.

Ex. 4.3.8 : *Classify ordinary points, singular points regular singular points and irregular singular points (if exist) of the differential equation* $y''+xy'=0$.

GTU : Summer-19, Maths-2, Marks 4

Sol. : Here, $y'' + xy' = 0$ is given.

First, we compare with standard equation.

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = x \text{ and } Q(x) = 1$$

Here, both are analytic everywhere.

$\therefore$ All points are ordinary.

Ex. 4.3.9 : *Classify the points of the equation* $xy'' + y' = 0$.

Sol. : Here, $xy'' + y' = 0$ is given.

$\therefore$ First, we write (Divide by x)

$$y'' + \frac{1}{x} y' = 0$$

Compare with $y'' + (Px) \cdot y' + Q(x) \cdot y = 0$

$\therefore P(x) = \frac{1}{x} \qquad Q(x) = 0$

P(x) is not analytic at $x = 0$.

$\therefore$ $x = 0$ is a singular points and others are ordinary points.

Now we check $x = 0$ is regular and irregular singular point.

For $\qquad x = 0$

$$(x-0)P(x) = \frac{(x-0)}{x} = 1$$

$$(x-0)^2 Q(x) = 0 = 0$$

$\therefore$ $(x - 0)$ P(x) and $(x-0)^2$ Q(x) are both analytic.

$\therefore$ $x = 0$ is regular singular point.

Ex. 4.3.10 : *Determine the singular points of the differential equation* $x(x+1)^2 y'' + (2x-1)y' + x^2 y = 0$ *and classify them as regular or irregular.*

GTU : Winter-19, AEM, Marks 3

Sol. : Here, $x(x+1)^2 y'' + (2x-1)y' + x^2 y = 0$ is given.

First, we write (Divide by $x(x+1)^2$)

$$\therefore \quad y'' + \frac{(2x-1)}{x(x+1)^2} y' + \frac{x}{(x+1)} y = 0$$

Compare with standard form.

$$\therefore \quad y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = \frac{2x-1}{x(x+1)^2}, \; Q(x) = \frac{x}{x+1}$$

Here, P(x) is not analytic at $x = 0$ and $x = -1$ and Q(x) is not analytic at $x = -1$.

$\therefore$ $x = 0$ and $x = -1$ are singular points. Others points are ordinary.

Now, we check $x = 0$ and $x = -1$ is regular or irregular.

For $x = 0$

$$(x-0)P(X) = \frac{(x-0)(2x-1)}{x(x+1)^2} = \frac{(2x-1)}{(x+1)^2}$$

$$(x-0)^2 Q(x) = \frac{(x-0)^2 \cdot x}{x+1} = \frac{x^3}{x+1}$$

Here, $(x - 0)$ P(x) and $(x-0)^2 Q(x)$ also analytic at $x = 0$.

$\therefore$ $x = 0$ is regular singular point.

For $x = -1$

$$(x+1)P(x) = \frac{(x+1)(2x-1)}{x(x+1)^2} = \frac{(2x-1)}{x(x+1)}$$

$$(x+1)^2 Q(x) = \frac{(x+1)^2 \cdot x}{(x+1)} = (x+1) \cdot x$$

Here, $(x + 1)$ P(x) is not analytic at $x = -1$.

$\therefore$ $x = -1$ is irregular singular point.

Ex. 4.3.11 : *Find the ordinary and singular points of the equation* $(1-x^2)y'' - 6xy' - 4y = 0$.

GTU : Winter-22, Maths-III, Marks 7

Sol. : Here, $x(1-x^2)y'' - 6xy' - 4y = 0$ is given.

First, we write (Divide by $(1-x^2)$)

$$y'' - \frac{6xy'}{(1-x^2)} - \frac{4}{(1-x^2)}y = 0$$

Compare with standard form.

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = \frac{-6x}{(1-x^2)} = \frac{-6x}{(1-x)(1+x)}$$

$$Q(x) = \frac{-4}{(1-x^2)} = \frac{-4}{(1-x)(1+x)}$$

Here, P(x) and Q(x) both not analytic at $x = 1$ and $x = -1$.

$\therefore$ $x = 1$ and $x = -1$ both singular points. Other points are ordinary.

4.4 Power Series Solution at Ordinary Point

GTU : Winter-18,19,21,22,23, Summer-18,19,20,21,22,23,24, Marks 7

Let us consider a linear 2^{nd} order ordinary differential equation with variable co-efficient.

$$a(x) \cdot \frac{d^2y}{dx^2} + b(x) \cdot \frac{dy}{dx} + c(x) \cdot y = 0 \quad \text{...(A)}$$

If $x = x_0$ is a ordinary points then we follow following steps.

1. Assume the solution of equation (A) is,

$$y = \sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + ... + a_n(x-x_0)^n + ...$$

If $x_0 = 0$ then

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + ... + a_n x^n + ...$$

2. Find derivative $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$
3. Put y, y′ and y″ in given equation.
4. Compairing both sides and find the value of the co-efficient $a_2, a_3, a_4, ...$ in terms of a_0 and a_1 (When equation is 2^{nd} order) (If equation is 1^{st} order then find co-efficient $a_1, a_2, a_3, ...$ in terms of a_0).
5. Put all values of the co-efficient $(a_2, a_3, ...)$ or $(a_1, a_2, a_3, ...)$ in assume solution
$$y = \sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + ...$$

which is required solution.

Ex. 4.4.1 : *Find the power series solution of* $y'' + y = 0$.

GTU : Summer-20, 24, AEM, Marks 7, Summer-19, Maths-3, Marks 7

Sol. : Here, $y'' + y = 0$ is given.

Let us compared given equation with standard form.

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$\therefore$ $$P(x) = 0 \text{ and } Q(x) = 1$$

$\therefore$ $P(x)$ and $Q(x)$ both analytic everywhere.

$\therefore$ All points are ordinary.

$\therefore$ $x = 0$ is a ordinary point.

$\therefore$ Let us assume series solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \ldots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \ldots$$

Put y'' and y in given equation.

$\therefore$ $$y''+y = 0$$

$$(2a_2+6a_3x+12a_4x^2+20a_5x^3+\ldots)\ (a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+\ldots) = 0$$

$$(2a_2+a_0)+(6a_3+a_1)x+(12a_4+a_2)x^2+(20a_5+a_3)x^3+\ldots = 0+0\cdot x+0x^2+0x^3+\ldots$$

Compare co-efficient of $x^0, x^1, x^2, x^3, \ldots$ both side.

$\therefore$ $$2a_2 + a_0 = 0 \Rightarrow a_2 = -\frac{a_0}{2}$$

$$6a_3 + a_1 = 0 \Rightarrow a_3 = -\frac{a_1}{6}$$

$$12a_4 + a_2 = 0 \Rightarrow 12a_4 = -\frac{a_0}{2} = 0$$

$$\Rightarrow 12a_4 = \frac{a_0}{2}$$

$$\Rightarrow a_4 = \frac{a_0}{24}$$

$$20a_5 + a_3 = 0 \Rightarrow 20a_5 = -a_3$$

$$\Rightarrow 20a_5 = +\frac{a_1}{6}$$

$$\Rightarrow a_5 = \frac{a_1}{120}$$

Put a_2, a_3, a_4 and a_5 in assume solution,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$= a_0 + a_1x - \frac{a_0}{2}x^2 - \frac{a_1}{6}x^3 + \frac{a_0}{24}x^4 + \frac{a_1}{120}x^5 + \ldots$$

$$y = a_0\left(1-\frac{1}{2}x^2+\frac{1}{24}x^4+\ldots\right)+a_1\left(x-\frac{1}{6}x^3+\frac{1}{120}x^5+\ldots\right)$$

Ex. 4.4.2 : *Solve $y' = 2xy$ using power series.*

GTU : Winter-19, AEM, Marks 4, Summer-20, Maths-2, Winter-22, Maths-3, Marks 7

Sol : Here, $y' = 2xy$ is given.

Compare with standard form.

$$y'' + P(x) \cdot y' + Q(x) \cdot y = 0$$

$$\therefore \quad P(x) = 1 \text{ and } Q(x) = 2x$$

$\therefore$ P(x) and Q(x) both analytic everywhere.

$\therefore$ All points are ordinary.

$\therefore$ $x = 0$ is also ordinary point.

$\therefore$ Let us assume solution of given equation is,

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$\therefore \quad y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots$$

put y' and y in given equation.

$$\therefore \quad y' = 2xy$$

$$\therefore \quad y' - 2xy = 0$$

$$\therefore \quad (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots) - 2x(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) = 0$$

$$\therefore \quad (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots) - 2a_0 x - 2a_1 x^2 - 2a_2 x^3 - 2a_3 x^4 + \dots = 0$$

$$\therefore \quad a_1 + (2a_2 - 2a_0)x + (3a_3 - 2a_1)x^2 + (4a_4 - 2a_2)x^3 + \dots = 0 + 0 \cdot x + 0x^2 + \dots$$

Compare both sides of co-efficient of $x^0, x^1, x^2, x^3, \dots$

$$\therefore \quad a_1 = 0$$

$$\therefore \quad 2a_2 - 2a_0 = 0 \Rightarrow a_2 = a_0$$

$$\therefore \quad 3a_3 - 2a_1 = 0 \Rightarrow a_3 = 0$$

$$4a_4 - 2a_2 = 0 \Rightarrow 4a_4 = 2a_2$$

$$\therefore \quad 4a_4 = 2a_0$$

$$\therefore \quad a_4 = \frac{1}{2} a_0$$

Now, put a_1, a_2, a_3, a_4 in assume solution.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$$

$$= a_0 + 0 \cdot x + a_0 x^2 + 0 \cdot x^3 + \frac{1}{2} a_0 x^4 + \dots$$

$$\therefore \quad y = a_0 \left(1 + x^2 + \frac{x^4}{2} + \dots \right)$$

Ex. 4.4.3 : *Solve $y'' + xy = 0$ using power series.*

GTU : Winter-19, 22, Maths-2, Summer-20, 22, 23, Maths-3, Winter-21, 23, AEM, Marks 7

Sol. : Here, $y'' + xy = 0$ is given.

Compare with $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$

$\therefore$ $P(x) = 0$ and $Q(x) = x$

$\therefore$ Here, $P(x)$ and $Q(x)$ both analytic everywhere.

$\therefore$ All points are ordinary.

$\therefore$ $x = 0$ is a ordinary points.

$\therefore$ Let us assume solution of given equation.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \ldots$$

$$y'' = 2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots$$

Now, put y and y″ in given equation.

$$y'' + xy = 0$$

$$(2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots) + (a_0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \ldots) = 0$$

$$\therefore \; 2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots + a_0 x + a_1 x^2 + a_2 x^3 + a_3 x^4 + \ldots = 0 + 0x + 0x^2 + 0x^3 + \ldots$$

Compare both sides of co-efficient of $x^0, x^1, x^2, \ldots$

$$\therefore \; 2a_2 + (6a_3 + a_0)x + (12a_4 + a_1)x^2 + (20a_5 + a_2)x^3 + \ldots = 0 + 0x + 0x^2 + 0x^3 + \ldots$$

$$\therefore \quad 2a_2 = 0 \Rightarrow a_2 = 0$$

$$6a_3 + a_0 = 0 \Rightarrow a_3 = -\frac{a_0}{6}$$

$$12a_4 + a_1 = 0 \Rightarrow a_4 = -\frac{a_1}{12}$$

$$20a_5 + a_2 = 0 \Rightarrow a_5 = 0$$

Put a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$\therefore \quad y = a_0 + a_1 x + 0 \cdot x^2 - \frac{a_0}{6} x^3 - \frac{a_1}{12} x^4 + 0.x^5 + \ldots$$

$$\therefore \quad y = a_0\left(1 - \frac{x^3}{6} + \ldots\right) + a_1\left(x - \frac{x^4}{12} + \ldots\right)$$

Ex. 4.4.4 : *Find the series solution of $y'' - y' = 0$.*

GTU : Winter-23, Maths-2, Marks 4

Sol. : Here, $y'' - y' = 0$ is given.

Compare with $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$

$$\therefore \quad P(x) = -1, Q(x) = 0$$

Both P(x) and Q(x) are analytic everywhere.

$\therefore$ All points are ordinary.

$\therefore$ $x = 0$ is ordinary point.

Now, let us assume the solution of

$$y''-y' = 0 \text{ is,}$$

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$\therefore \quad y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \ldots$$

$$\therefore \quad y'' = 2a_2 + 6a_3 x + 12a_4 x^4 + 20a_5 x^3 + \ldots$$

Now, put y' and y'' in given equation.

$$\therefore \quad y''-y' = 0$$

$$\therefore \quad (2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots) - (a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \ldots) = 0$$

$$\therefore \quad (2a_2 - a_1) + (6a_3 - 2a_2)x + (12a_4 - 3a_3)x^2 + (20a_5 - 4a_4)x^3 + \ldots = 0 + 0x + 0x^2 + 0x^3 + \ldots$$

Compare co-efficient both sides of $x^0, x^1, x^2, x^3, \ldots$

$$\therefore \quad 2a_2 - a_1 = 0 \Rightarrow a_2 = \frac{a_1}{2}$$

$$6a_3 - 2a_2 = 0 \Rightarrow 6a_3 = 2a_2$$

$$\Rightarrow a_3 = \frac{1}{3}a_2 = \frac{a_1}{6}$$

$$12a_4 - 3a_3 = 0 \Rightarrow a_4 = +\frac{3}{12}a_3 = +\frac{1}{4}a_3$$

$$\therefore \quad a_4 = +\frac{1}{4}\cdot\frac{a_1}{6} = +\frac{a_1}{24}$$

$$20a_5 - 4a_4 = 0 \Rightarrow a_5 = +\frac{4}{20}a_4 = +\frac{1}{5}a_4$$

$$\therefore \quad a_5 = +\frac{1}{5}\left(+\frac{a_1}{24}\right) = \frac{a_1}{120}$$

Put a_2, a_3, a_4 and a_5 in the solution

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$\therefore \quad y = a_0 + a_1 x + \frac{a_1}{2}x^2 + \frac{a_1}{6}x^3 + \frac{a_1}{24}x^4 + \frac{a_1}{120}x^5 + \ldots$$

$$\therefore \quad y = a_0 + a_1\left(x + \frac{x_2}{2} + \frac{x_3}{6} + \frac{x_4}{20} + \frac{x_5}{120} + \ldots\right)$$

Ex. 4.4.5 : *Find the power series solution of $y' + 2xy = 0$ in the power of x.* **GTU : Winter-23, Maths-2, Marks 4**

Sol. : Here, $y' + 2xy = 0$

Compare with standard form.

$$y'+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$P(x) = 1 \text{ and } Q(x) = 2x$$

$\therefore$ Both P(x) and Q(x) analytic everywhere.

$\therefore$ All points are ordinary.

$\therefore x = 0$ is ordinary point.

Let use take the solution,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$

$$\therefore \quad y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...$$

put, y and y' in given equation.

$$y' + 2xy = 0$$

$$\therefore \quad (a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...) + 2x(a_0 + a_1x + a_2x^2 + a_3x^3 + ...) = 0$$

$$\therefore \quad (a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...) + (2a_0x + 2a_1x^2 + 2a_2x^3 + 2a_3x^4 + ...) = 0 + 0x + 0x^2 + 0x^3 + ...$$

$$\therefore \quad (a_1) + (2a_2 + 2a_0)x + (3a_3 + 2a_1)x^2 + (4a_4 + 2a_2)x^3 + ... = 0 + 0x + 0x^2 + 0x^3 + ...$$

Compare both side co-efficients of $x^0, x^1, x^2, x^3, ...$

$$\therefore \quad a_1 = 0$$

$$2a_2 + 2a_0 = 0 \Rightarrow a_2 = -a_0$$

$$3a_3 + 2a_1 = 0 \Rightarrow a_3 = 0$$

$$4a_4 + 2a_2 = 0 \Rightarrow a_4 = \frac{-a_2}{2} = \frac{a_0}{2}$$

Put a_1, a_2, a_3, a_4 in assume solution.

$$\therefore \quad y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$

$$= a_0 + 0\cdot x - a_0x^2 + 0\cdot x^3 + \frac{a_0}{2}x^4 + ...$$

$$\therefore \quad y = a_0\left(1 - x^2 + \frac{x^4}{2} - + ...\right)$$

Ex. 4.4.6 : *Find the series solution of $(1+x^2)y'' + xy' - 9y = 0$ near the ordinary point $x = 0$.*

GTU : Winter-21, Summer-24, Maths-2, Winter-21, Maths-3, Marks 7

Sol. : Here, $(1+x^2)y'' + xy' - 9y = 0$ is given and $x = 0$ is ordinary point is given.

Let us assume the solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + \dots$$

Put, y, y′ and y″ in given equation.

$$\therefore \quad (1+x^2)y'' + xy' - 9y = 0$$

$$\therefore \quad y'' + x^2y'' + xy' - 9y = 0$$

$$(2a_2 + 6a_3x + 12a_4x^2 + \dots) + x^2(2a_2 + 6a_3x + 12a_4x^2 + \dots)$$
$$+ x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots) - 9(a_0 + a_1x + a_2x^2 + a_3x^3 + \dots) = 0$$

$$\therefore (2a_2 + 6a_3x + 12a_4x^2 + \dots) + (2a_2x^2 + 6a_3x^3 + 12a_4x^4 + \dots)$$
$$+ (a_1x + 2a_2x^2 + 3a_3x^3 + 4a_4x^4 + \dots) - (9a_0 + 9a_1x + 9a_2x^2 + 9a_3x^3 + \dots) = 0$$

$$\therefore (2a_2 - 9a_0) + (6a_3 + a_1 - 9a_1)x + (12a_4 + 2a_2 + 2a_2 - 9a_2)x^2 + \dots = 0 + 0\cdot x + 0\cdot x^2 + \dots$$

Compare co-efficient both sides of $x^0, x^1, x^2, \dots$

$$\therefore \quad 2a_2 - 9a_0 = 0 \Rightarrow a_2 = \frac{9}{2}a_0$$

$$6a_3 + a_1 - 9a_1 = 0 \Rightarrow 6a_3 - 8a_1 = 0 \Rightarrow a_3 = \frac{4}{3}a_1$$

$$12a_4 + 4a_2 - 9a_2 = 0 \Rightarrow 12a_4 = +5a_2$$

$$\Rightarrow a_4 = \frac{5}{12}a_2$$

$$= \frac{5}{12}\cdot\frac{9}{2}a_0$$

$$\therefore \quad a_4 = \frac{45}{24}a_0$$

Put a_2, a_3, a_4 in assume solution.

$$\therefore \quad y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

$$= a_0 + a_1x + \frac{9}{2}a_0x^2 + \frac{4}{3}a_1x^3 + \frac{45}{4}a_0x^4 + \dots$$

$$\therefore \quad y = a_0\left(1 + \frac{9}{2}x^2 + \frac{45}{24}x^4 + \dots\right) + a_1\left(x + \frac{4}{3}x^3 + \dots\right)$$

Ex. 4.4.7 : *Find the power series solution of $y'' - 25y = 0$ near an ordinary point $x = 0$.*

GTU : Summer-23, Maths-2, Marks 7

Sol. : Here, $y'' - 25y = 0$ is given.

Here also $x = 0$ is ordinary point.

∴ Let us assume the solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots$$

$$\therefore \quad y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \dots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \dots$$

Now put y and y″ in given equation.

$$y''-25y = 0$$

$$(2a_2+6a_3x+12a_4x^2+20a_5x^3+\dots)-25(a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+\dots)=0$$

$$\therefore (2a_2+6a_3x+12a_4x^2+20a_5x^3+\dots)-(25a_0+25a_1x+25a_2x^2+25a_3x^3+\dots)=0+0x+0x^2+0x^3+\dots$$

$$\therefore (2a_2-25a_0)+(6a_3-25a_1)x+(12a_4-25a_2)x^2+(20a_5-25a_3)x^3+\dots=0+0x+0x^2+0x^3+\dots$$

Compare both sides co-efficient of $x^0, x^1, x^2, x^3, \dots$

$$\therefore \quad 2a_2-25a_0=0 \Rightarrow a_2=\frac{25}{2}a_0$$

$$6a_3-25a_1=0 \Rightarrow a_3=\frac{25}{6}a_1$$

$$12a_4-25a_2=0 \Rightarrow a_4=\frac{25}{12}a_2 = \frac{25}{12}\times\frac{25}{2}a_0 = \frac{625}{24}a_0$$

$$20a_5-25a_3 = 0 \Rightarrow a_5=\frac{25}{20}a_3 = \frac{5}{4}a_3=\frac{5}{4}\times\frac{25}{6}a_1 = \frac{125}{24}a_1$$

Put a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\dots$$

$$\therefore \quad y = a_0+a_1x+\frac{25}{2}a_0x^2+\frac{25}{6}a_1x^3+\frac{625}{24}a_0x^4+\frac{125}{24}a_1x^5+\dots$$

$$\therefore \quad y = a_0\left(1+\frac{25}{2}x^2+\frac{625}{24}x^4+\dots\right)+a_1\left(x+\frac{25}{6}x^3+\frac{125}{24}x^5+\dots\right)$$

Ex. 4.4.8 : *Find the power series solution of the equation $(x^2+1)y''+xy'-xy=0$ in the power of x.*

GTU : Summer-22, Maths-2, Summer-22, AEM, Marks 7

Sol. : Here, $(x^2+1)y''+xy'-xy=0$ is given.

First we write (Divide by x^2+1)

$$y''+\frac{x}{x^2+1}y'-\frac{x}{x^2+1}y = 0$$

Compare with standard equation,

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = \frac{x}{x^2+1}, \quad Q(x)=\frac{-x}{x^2+1}$$

Here, P(x) and Q(x) analytic for all $x \in R$.

$\therefore$ All points are ordinary.

$\therefore$ $x = 0$ is also ordinary.

Let us assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$
$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...$$
$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + ...$$

Put, y, y′ and y″ in given equation,

$$(x^2+1)y'' + xy' - xy = 0$$

$$\therefore (x^2+1)(2a_2 + 6a_3x + 12a_4x^2 + ...) + x(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + ...)$$
$$- x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...) = 0$$

$$\therefore (2a_2x^2 + 6a_3x^3 + 12a_4x^4 + ...) + (2a_2 + 6a_3x + 12a_4x^2 + ...) + (a_1x + 2a_2x^2 + 3a_3x^3 + 4a_4x^4 + ...)$$
$$- (a_0x + a_1x^2 + a_2x^3 + a_3x^4 + ...) = 0$$

$$\therefore 2a_2 + (6a_3 + a_1 - a_0)x + (2a_2 + 12a_4 + 2a_2 - a_1)x^2 + ... = 0 + 0x + 0x^2 + ...$$

Compare both side co-efficient of $x^0, x^1, x^2, ...$

$$\therefore \quad 2a_2 = 0 \Rightarrow a_2 = 0$$
$$6a_3 + a_1 - a_0 = 0 \Rightarrow a_3 = \frac{a_0 - a_1}{6}$$
$$2a_2 + 12a_4 + 2a_2 - a_1 = 0 \Rightarrow 12a_4 + 4a_2 - a_1 = 0$$
$$\Rightarrow 12a_4 = a_1$$
$$\Rightarrow a_4 = \frac{a_1}{12}$$

Put a_2, a_3, a_4 in assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...$$
$$= a_0 + a_1x + 0x^2 + \left(\frac{a_0 - a_1}{6}\right)x^3 + \frac{a_1}{12}x^4 + ...$$

$$\therefore \quad y = a_0\left(1 + \frac{1}{6}x^3 + ...\right) + a_1\left(x - \frac{x^3}{6} + \frac{x^4}{12} + ...\right)$$

Ex. 4.4.9 : *Find the power series solution of the differential equation $y'' - xy = 0$ near an ordinary point $x = 0$.*

GTU : Summer-19, Maths-2, Winter-19, Maths-3, Marks 7

Sol. : Here, $y'' - xy = 0$ is given.

Here, $x = 0$ is a ordinary points.

Let us assume the solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + ...$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + ...$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...$$

Put y, y′ and y″ in given equation,

$$y'' - xy = 0$$

$$(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...) - x(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...) = 0$$

$$\therefore \ (2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...) - (a_0x + a_1x^2 + a_2x^3 + a_3x^4 + a_4x^5 + ...) = 0$$

$$\therefore \ 2a_2 + (6a_3 - a_0)x + (12a_4 - a_1)x^2 + (20a_5 - a_2)x^3 + ... = 0 + 0x + 0x^2 + 0x^3 + ...$$

Compare both side co-efficient of $x^0, x^1, x^2, x^3, ...$

$$2a_2 = 0 \Rightarrow a_2 = 0$$

$$\therefore \quad 6a_3 - a_0 = 0 \Rightarrow a_3 = \frac{a_0}{6}$$

$$12a_4 = a_1 = 0 \Rightarrow a_4 = \frac{a_1}{12}$$

$$20a_5 - a_2 = 0 \Rightarrow a_5 = 0$$

Now, Put, a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4\ x^4 + ...$$

$$y = a_0 + a_1x + 0 \cdot x^2 + \frac{a_0}{6}x^3 + \frac{a_1}{12}x^4 + ...$$

$$= a_0\left(1 + \frac{x^3}{6} + ...\right) + a_1\left(x + \frac{x^4}{12} + ...\right)$$

Ex. 4.4.10 : *Find the power series solution of $y'' + x^2y = 0$.*

GTU : Summer-18, Winter-22, AEM, Winter-19, AEM, Marks 4, Summer-18, Winter-23, Maths-3, Marks 7

Sol. : Here, $y'' + x^2y = 0$ is given.

$x = 0$ is a ordinary point.

Let us assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + ...$$

$$y' = a_1 + 2a_2x + 2a_3x^2 + 4a_4x^3 + 5a_5x^4 + ...$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...$$

Put y and y″ in given equation,

$$y'' + x^2y = 0$$

$$(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...) + x^2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + ...) = 0$$

$$(2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + ...) + (a_0x^2 + a_1x^3 + a_2x^4 + a_3x^5 + a_4x^6 + ...) = 0$$

$$\therefore \quad 2a_2 + 6a_3x + (12a_4 + a_0)x^2 + (20a_5 + a_1)x^3 + \ldots = 0 + 0x + 0x^2 + 0x^3 + \ldots$$

Compare both side co-efficient of $x^0, x^1, x^2, x^3, \ldots$

$$\therefore \quad 2a_2 = 0 \Rightarrow a_2 = 0$$

$$6a_3 = 0 \Rightarrow a_3 = 0$$

$$12a_4 + a_0 = 0 \Rightarrow a_4 = \frac{-a_0}{12}$$

$$20a_5 + a_1 = 0 \Rightarrow a_5 = \frac{-a_1}{20}$$

Put a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$= a_0 + a_1x + 0 \cdot x^2 + 0 \cdot x^3 - \frac{a_0}{12}x^4 - \frac{a_1}{20}x^5 + \ldots$$

$$= a_0\left(1 - \frac{x^4}{12} + \ldots\right) + a_1\left(x - \frac{x^5}{20} + \ldots\right)$$

Ex. 4.4.11 : *Solve in series the equation $y' = 3x^2y$.* **GTU : Winter-18, AEM, Marks**

Sol. : Here, $y' = 3x^2y$ is given and $x = 0$ is ordinary point.

$\therefore$ Let us assume solution of given equation,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \ldots$$

Now, put y and y′ in given equation,

$$y' = 3x^2y \Rightarrow y' - 3x^2y = 0$$

$$(a_1 + 2a_2x + 3a_3x^2 + 4a^4x^3 + 5a_5x^4 + \ldots) - 3x^2(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \ldots) = 0$$

$$a_1 + 2a_2x + (3a_3 - 3a_0)x^2 + (4a_4 - 3a_1)x^3 + (5a_5 - 3a_2)x^4 + \ldots = 0 + 0 \cdot x + 0x^2 + \ldots$$

Compare co-efficient of $x^0, x^1, x^2, x^3, x^4, \ldots$

$$\therefore \quad a_1 = 0, \; 2a_2 = 0 \Rightarrow a_2 = 0$$

$$3a_3 - 3a_0 = 0 \Rightarrow a_3 = a_0$$

$$4a_4 - 3a_1 = 0 \Rightarrow a_4 = 0$$

$$5a_5 - 3a_2 = 0 \Rightarrow a_5 = 0$$

Put, a_1, a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$\therefore \quad y = a_0 + 0 \cdot x + 0 \cdot x^2 + a_0x^3 + 0 \cdot x^4 + 0 \cdot x^5 + \ldots$$

$$\therefore \quad y = a_0(1 + x^3 + \ldots)$$

Ex. 4.4.12 : *Find the power series solution of* $(x^2+1)\dfrac{d^2y}{dx^2}+2x\dfrac{dy}{dx}+xy=0.$

GTU : Summer-23, AEM, Marks 7

Sol. : Here, $(x^2+1)y''+2xy'+xy=0$ is given.

First, we write (Divide by x^2+1)

$$y''+\frac{2x}{x^2+1}y'+\frac{x}{x^2+1}y = 0$$

$$\therefore \quad P(x) = \frac{2x}{x^2+1} \quad Q(x) = \frac{x}{x^2+1}$$

$\therefore$ P(x) and Q(x) both analytic for all value of $x \in R$.

$\therefore$ All points are ordinary.

$\therefore$ $x=0$ is ordinary point.

Let us consider the solution of given equation.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+a_6x^6+...$$

$$\therefore \quad y' = a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+6a_6x^5+...$$

$$y'' = 2a_2+6a_3x+12a_4x^2+20a_5x^3+30a_6x^4+...$$

Put, y, y' and y'' in given equation,

$$\therefore (x^2+1)y''+2xy'+xy = 0$$

$$\therefore x^2y''+y''+2xy'+xy = 0$$

$$\therefore x^2(2a_2+6a_3x+12a_4x^2+20a_5x^3+30a_6x^4+...)+(2a_2+6a_3x+12a_4x^2+20a_5x^3+30a_6x^4+...)$$
$$+2x(a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+...)+x(a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+...) = 0$$

$$\therefore (2a_2x^2+6a_3x^3+12a_4x^4+20a_5x^5+...)+(2a_2+6a_3x+12a_4x^2+20a_5x^3+30a_6x^4+...)$$
$$+(2a_1x+4a_2x^2+6a_3x^3+8a_4x^4+10a_5x^5+...)+(a_0x+a_1x^2+a_2x^3+a_3x^4+a_4x^5+...) = 0$$

$$\therefore 2a_2+(6a_3+2a_1+a_0)x+(2a_2+12a_4+4a_2+a_1)x^2+(6a_3+20a_5+6a_3+a_2)x^3$$
$$+(12a_4+30a_6+8a_4+a_3)x^4+... = 0+0x+0x^2+...$$

Compare co-efficient both sides.

$$\therefore \quad 2a_2 = 0 \Rightarrow a_2 = 0$$

$$6a_3+2a_1+a_0 = 0 \Rightarrow a_3 = \frac{-(2a_1+a_0)}{6}$$

$$a_3 = \frac{-a_1}{3}-\frac{a_0}{6}$$

$$2a_2+12a_4+4a_2+a_1 = 0$$

$$\Rightarrow \quad 12a_4+6a_2+a_1 = 0$$

$$\Rightarrow \quad 12a_4 = -6a_2-a_1$$

$$\Rightarrow \quad a_4 = -\frac{1}{2}a_2 - \frac{1}{2}a_1 = -\frac{1}{12}a_1$$

$$6a_3 + 20a_5 + 6a_3 + a_2 = 0$$

$$\Rightarrow \quad 20a_5 + 12a_3 = 0$$

$$\Rightarrow \quad a_5 = \frac{-12}{20}a_3 = \frac{-3}{5}a_3$$

$$= \frac{-3}{5}\left(\frac{-a_1}{3} - \frac{a_0}{6}\right)$$

$$= \frac{a_1}{5} + \frac{a_0}{2}$$

$$12a_4 + 30a_6 + 8a_4 + a_3 = 0$$

$$\Rightarrow \quad 30a_6 + 20a_4 + a_3 = 0$$

$$\Rightarrow \quad 30a_6 = -20a_4 - a_3$$

$$\Rightarrow \quad a_6 = \frac{-2}{3}a_4 - \frac{1}{30}a_3$$

$$= \frac{-2}{3}\left(-\frac{1}{12}a_1\right) - \frac{1}{30}\left(\frac{-a_1}{3} - \frac{a_0}{6}\right)$$

$$= \frac{1}{18}a_1 + \frac{a_1}{90} + \frac{a_0}{180}$$

$$= \frac{54}{81}a_1 + \frac{a_0}{180}$$

Put a_2, a_3, a_4, a_5, a_6 in assume solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \ldots$$

$$\therefore \quad y = a_0 + a_1x + 0 + \left(-\frac{a_1}{3} - \frac{a_0}{6}\right)x^3 + \left(-\frac{1}{12}a_1\right)x^4 + \left(\frac{a_1}{5} + \frac{a_0}{2}\right)x^5 + \left(\frac{54}{81}a_1 + \frac{a_0}{180}\right)x^6 + \ldots$$

$$\therefore \quad y = a_0\left(1 - \frac{1}{6}x^3 + \frac{1}{2}x^5 + \frac{1}{180}x^6 + \ldots\right) + a_1\left(x - \frac{x^3}{3} - \frac{x^4}{12} + \frac{x^5}{5} + \frac{54}{81}x^6 + \ldots\right)$$

Ex. 4.4.13 : *Find the series solution of $y'' + xy' + y = 0$ near the ordinary point $x = 0$.* **GTU : Summer-20, AEM, Marks 7**

Sol. : Here, $y'' + xy' + y = 0$ is given and also $x = 0$ is a ordinary point.

$\therefore$ Let us assume the solution.

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$$

$$y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 5a_5x^4 + \ldots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + 20a_5x^3 + \ldots$$

Put y, y′ and y″ in given equation,

$$y'' + xy' + y = 0$$

$\therefore\ (2a_2+6a_3x+12a_4x^2+20a_5x^3+...)+(a_1x+2a_2x^2+3a_3x^3+4a_4x^4+...)$

$+(a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+...)=0$

$\therefore\ (2a_2+a_0)+(6a_3+a_1+a_1)x+(12a_4+2a_2+a_2)x^2+(20a_5+3a_3+a_3)x^3+...=0+0x+0x^2+...$

Compare co-efficient both sides,

$$2a_2+a_0 = 0 \Rightarrow a_2=-\frac{a_0}{2}$$

$$6a_3+2a_2 = 0 \Rightarrow a_3=-\frac{2a_1}{6}=-\frac{a_1}{3}$$

$$12a_4+3a_2 = 0 \Rightarrow a_4=-\frac{1}{4}a_2=\frac{a_0}{8}$$

$$20a_5+4a_3 = 0 \Rightarrow a_5=-\frac{1}{5}a_3=\frac{a_1}{15}$$

Put a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+...$$

$$= a_0+a_1x-\frac{a_0}{2}x^2-\frac{a_1}{3}x^3+\frac{a_0}{8}x^4+\frac{a_1}{15}x^5+...$$

$$\therefore\quad y = a_0\left(1-\frac{x^2}{2}+\frac{x^4}{8}+...\right)+a_1\left(x-\frac{x^3}{3}+...\right)$$

Ex. 4.4.14 : *Find series solution of the differential equation $(1+x^2)y''+xy'-y=0$.* **GTU : Winter-18, Maths-3, Marks 7**

Sol. : Here, $(1+x^2)y''+xy'-y=0$ is given

First, we write $(1+x^2)$ divide 1

$$y''+\frac{x}{(1+x^2)}y'-\frac{1}{(1+x^2)}y = 0$$

Compare with standard form.

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore\quad P(x) = \frac{x}{1+x^2},\quad Q(x)=-\frac{1}{(1+x^2)}$$

Here, Both P(x) and Q(x) are analytic for all values of x, $x \in R$

$\therefore$ All points are ordinary.

$\therefore$ $x=0$ is also ordinary.

Let us assume the solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+...$$

$$y' = a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+...$$

$$y'' = 2a_2+6a_3x+12a_4x^2+20a_5x^3+...$$

Put y, y′ and y″ in given equation,

$$(1+x^2)y''+xy'-y = 0$$

$$(1+x^2)(2a_2+6a_3x+12a_4x^2+20a_5x^3+\ldots)+x(a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+\ldots)$$

$$-(a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\ldots)=0$$

$$\therefore\ (2a_2+6a_3x+12a_4x^2+20a_5x^3+\ldots)+(2a_2x^2+6a_3x^3+12a_4x^4+20a_5x^5+\ldots)$$

$$+(a_1x+2a_2x^2+3a_3x^3+4a_4x^4+5a_5x^5+\ldots)-(a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\ldots)=0$$

$$\therefore\ (2a_2-a_0)+(6a_3+a_1-a_1)x+(12a_4+2a_2+2a_2-a_2)x^2$$

$$+(20a_5+6a_3+3a_3-a_3)x^3+\ldots = 0+0x+0x^2+0x^3+\ldots$$

Compare co-efficient both sides,

$$\therefore\quad 2a_2-a_0 = 0 \Rightarrow a_2=\frac{a_0}{2}$$

$$6a_3+a_1-a_1 = 0 \Rightarrow a_3=0$$

$$12a_4+4a_2-a_2 = 0 \Rightarrow 12a_4+3a_2=0$$

$$\Rightarrow a_4 = -\frac{3}{12}a_2 = -\frac{1}{4}\left(\frac{a_0}{2}\right)$$

$$= -\frac{a_0}{8}$$

$$20a_5+9a_3-a_3 = 0 \Rightarrow 20a_5+8a_3=0$$

$$\Rightarrow a_5=0$$

Put a_2, a_3, a_4 and a_5 in assume solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\ldots$$

$$y = a_0+a_1x+\frac{a_0}{2}x^2+0x^3-\frac{a_0}{8}x^4+0x^5+\ldots$$

$$y = a_0\left(1+\frac{x^2}{2}-\frac{x^4}{8}+\ldots\right)+a_1x$$

Ex. 4.4.15 : *Find the power series solution of the equation* $\frac{d^2y}{dx^2}-\frac{2dy}{dx}=0.$

GTU : Summer-24, Maths-3, Marks 7

Sol. : Here, $y''-2y'=0$ is given.

Here, $x=0$ is ordinary point.

Let us assume the solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\ldots$$

$$y' = a_1+2a_2x+3a_3x^2+4a_4x^3+5a_5x^4+\ldots$$

$$y'' = 2a_2+6a_3x+12a_4x^2+20a_5x^3+\ldots$$

Put y' and y'' in given equation,

$$y''-2y' = 0$$

$$\therefore (2a_2+6a_3x+12a_4x^2+20a_5x^3+\ldots)-(2a_1+4a_2x+6a_3x^2+8a_4x^3+10a_5x^4+\ldots)=0$$

$$\therefore (2a_2-2a_1)+(6a_3-4a_2)x+(12a_4-6a_3)x^2+(20a_5-8a_4)x^3+\ldots=0+0x+0x^2+0x^3+\ldots$$

Compare co-efficient both sides,

$$2a_2-2a_1 = 0 \Rightarrow a_2 = a_1$$

$$6a_3-4a_2 = 0 \Rightarrow a_3=\frac{2}{3}a_2=\frac{2}{3}a_1$$

$$12a_4-6a_3 = 0 \Rightarrow a_4=\frac{1}{2}a_3=\frac{1}{2}\left(\frac{2}{3}\right)a_1=\frac{1}{3}a_1$$

$$20a_5-8a_4 = 0 \Rightarrow a_5=\frac{2}{5}a_4=\frac{2}{5}\left(\frac{1}{3}a_1\right)=\frac{2}{15}a_1$$

$\therefore$ Put a_2, a_3, a_4 and a_5 in assume solution.

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+a_5x^5+\ldots$$

$$y = a_0+a_1x+a_1x^2+\frac{2}{3}a_1x^3+\frac{1}{3}a_1x^4+\frac{2}{15}a_1x^5+\ldots$$

$$\therefore \quad y = a_0+a_1\left(a+x^2+\frac{2}{3}x^3+\frac{1}{3}x^4+\frac{2}{15}x^5+\ldots\right)$$

Ex. 4.4.16 : *Solve the equation by series method $(x-2)y''-x^2y'+9y=0$ about $x=0$.*

GTU : Summer-20, Maths-3, Marks 7

Sol. : Here, $(x-2)y''-x^2y'+9y=0$ is given.

First we write (Divide by x – 2)

$$\therefore y''-\frac{x^2}{x-2}y'+\frac{9}{x-2}y = 0$$

Compare with,

$$y''+P(x)\cdot y'+Q(x)\cdot y = 0$$

$$\therefore \quad P(x) = -\frac{x^2}{x-2} \quad Q(x)=\frac{9}{x-2}$$

Here, P(x) and Q(x) not analytic at x = 2.

$\therefore$ x = 2 is singular point. Others points are ordinary points.

$\therefore$ x = 0 is ordinary point.

$\therefore$ Let us assume solution,

$$y = a_0+a_1x+a_2x^2+a_3x^3+a_4x^4+\ldots$$

$$y' = a_1+2a_2x+3a_3x^2+4a_4x^3+\ldots$$

$$y'' = 2a_2 + 6a_3x + 12a_4x^2 + \ldots$$

Put y, y′ and y″ in given equation,

$$(x-2)y'' - x^2y' + 9y = 0$$

$$(2a_2x + 6a_3x^2 + 12a_4x^3 + \ldots) - (4a_2 + 12a_3 + 24a_4x^2 + \ldots)$$

$$-(a_1x^2 + 2a_2x^3 + 3a_3x^4 + 4a_4x^5 + \ldots) + (9a_0 + 9a_1x + 9a_2x^2 + 9a_3x^3 + 9a_4x^4 + \ldots) = 0$$

$$\therefore (4a_2 + 9a_0) + (2a_2 + 12a_3 + 9a_1)x + (6a_3 + 24a_4 - a_1 + 9a_2)x^2 + \ldots = 0 + 0x + 0x^2 + 0x^3 + \ldots$$

Compare co-efficient,

$$\therefore \quad 4a_2 + 9a_0 = 0 \Rightarrow a_2 = -\frac{9}{4}a_0$$

$$2a_2 + 12a_3 + 9a_1 = 0 \Rightarrow 12a_3 = -9a_1 - 2a_2$$

$$\Rightarrow a_3 = -\frac{9}{12}a_1 - \frac{2}{12}a_2 = -\frac{3}{4}a_1 - \frac{1}{6}a_2$$

$$= -\frac{3}{4}a_1 - \frac{1}{\cancel{6}}\left(-\frac{\cancel{9}^{3}}{4}\right)a_0$$

$$a_3 = -\frac{3}{4}a_1 + \frac{3}{8}a_0$$

$$6a_3 + 24a_4 - a_1 + 9a_2 = 0$$

$$\Rightarrow \quad 24a_4 = a_1 - 9a_2 - 6a_3$$

$$\Rightarrow \quad a_4 = \frac{1}{24}a_1 - \frac{9}{24}a_2 - \frac{6}{24}a_3$$

$$a_4 = \frac{1}{24}a_1 - \frac{3}{8}a_2 - \frac{1}{4}a_3 = \frac{1}{24}a_1 - \frac{1}{4}\left(-\frac{3}{4}a_1 + \frac{3}{8}a_0\right) - \frac{3}{8}\left(-\frac{9}{4}a_0\right)$$

$$= \frac{1}{24}a_1 + \frac{3}{16}a_1 - \frac{3}{32}a_0 + \frac{27}{32}a_0$$

$$= \frac{11}{48}a_1 + \frac{3}{4}a_0$$

Put a_2, a_3, a_4 in assume solution,

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \ldots$$

$$= a_0 + a_1x - \frac{9}{4}a_0x^2 + \left(-\frac{3}{4}a_1 + \frac{3}{8}a_0\right)x^3 + \left(\frac{11}{48}a_1 + \frac{3}{4}a_0\right)x^4 + \ldots$$

$$\therefore \quad y = a_0\left(1 - \frac{9}{4}x^2 + \frac{3}{8}x^3 + \frac{3}{4}x^4 + \ldots\right) + a_1\left(x - \frac{3}{4}x^3 + \frac{11}{48}x^4 + \ldots\right)$$

Ex. 4.4.17 : *Obtain the solution of $y'' - y = 0$.* **GTU : Summer-21, Maths-3, Marks**

Sol. : Here, $y'' - y = 0$ is given.

Here, $x = 0$ is ordinary points.

Let us assume solution.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + \ldots$$

$$y'' = 2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots$$

Put y and y″ in given equation,

$$y'' - y = 0$$

$$(2a_2 + 6a_3 x + 12a_4 x^2 + 20a_5 x^3 + \ldots) - (a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \ldots) = 0 + 0x + 0x^2 + 0x^3$$

Compare co-efficient both sides,

$$2a_2 - a_0 = 0 \Rightarrow a_2 = \frac{a_0}{2}$$

$$\therefore \quad 6a_3 - a_1 = 0 \Rightarrow a_3 = \frac{a_1}{6}$$

$$12a_4 - a_2 = 0 \Rightarrow a_4 = \frac{a_2}{12} = \frac{a_0}{24}$$

$$20a_5 - a_3 = 0 \Rightarrow a_5 = \frac{a_3}{20} = \frac{a_1}{120}$$

Put a_2, a_3, a_4, a_5 in assume solution.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + \ldots$$

$$= a_0 + a_1 x + \frac{a_0}{2} x^2 + \frac{a_1}{6} x^3 + \frac{a_0}{24} x^4 + \frac{a_1}{120} x^5 + \ldots$$

$$\therefore \quad y = a_0\left(1 + \frac{x^2}{2} + \frac{x^4}{24} + \ldots\right) + a_1\left(x + \frac{x^3}{6} + \frac{x^5}{120} + \ldots\right)$$

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